

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

บุณยพร อินทวงษ์ 663020582-2 เลขที่ 26

* มาใช้ ใช้ทฤษฎีบท แล้ว คำนวณ + เขียนต้นไม้

$$\text{Info}(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.904$$

age	p _i	n _i	I(p _i , n _i)
≤ 30	2	3	0.971
31...40	4	0	0
>40	3	2	0.971

$$\begin{aligned} \text{Info}_{\text{age}}(D) &= \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) \\ &= \frac{5}{14} \left[-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right] + \frac{4}{14} \left[-\frac{4}{4} \log_2\left(\frac{4}{4}\right) - \frac{0}{4} \log_2\left(\frac{0}{4}\right) \right] \\ &\quad + \frac{5}{14} \left[-\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right] \\ &= 0.347 + 0 + 0.347 = 0.694 \end{aligned}$$

income	p _i	n _i	I(p _i , n _i)
Low	3	1	0.811
Medium	4	2	0.918
high	2	2	1

$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) \\ &= \frac{4}{14} \left[-\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \right] + \frac{6}{14} \left[-\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \right] + \frac{4}{14} \left[-\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \right] \\ &= 0.232 + 0.394 + 0.296 = 0.912 \end{aligned}$$

student	p _i	n _i	I(p _i , n _i)
Yes	6	1	0.592
No	3	4	0.985

$$\begin{aligned} \text{Info}_{\text{student}}(D) &= \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4) \\ &= \frac{7}{14} \left[-\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) \right] + \frac{7}{14} \left[-\frac{3}{7} \log_2\left(\frac{3}{7}\right) - \frac{4}{7} \log_2\left(\frac{4}{7}\right) \right] \\ &= 0.296 + 0.493 = 0.789 \end{aligned}$$

credit_rating	p _i	n _i	I(p _i , n _i)
fair	5	2	0.863
excellent	4	3	0.985

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{7}{14} I(5,2) + \frac{7}{14} I(4,3) \\ &= \frac{7}{14} \left[-\frac{5}{7} \log_2\left(\frac{5}{7}\right) - \frac{2}{7} \log_2\left(\frac{2}{7}\right) \right] + \frac{7}{14} \left[-\frac{4}{7} \log_2\left(\frac{4}{7}\right) - \frac{3}{7} \log_2\left(\frac{3}{7}\right) \right] \\ &= 0.432 + 0.493 = 0.925 \end{aligned}$$

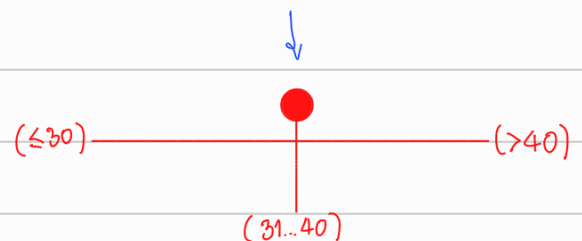
Gain(age) = 0.940 - 0.694 = 0.246 ✗ เลือก Age

Gain(income) = 0.940 - 0.912 = 0.028

Gain(student) = 0.940 - 0.789 = 0.151

Gain(credit_rating) = 0.940 - 0.925 = 0.015

Decision Tree



age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

$$\text{Info}(D) = I_{(2,3)} = -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) = 0.971$$

income	p_i	n_i	$I(p_i, n_i)$
Low	1	0	0
Medium	1	1	1
high	0	2	0

$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{1}{5} I_{(1,0)} + \frac{2}{5} I_{(1,1)} + \frac{2}{5} I_{(0,2)} \\ &= \frac{1}{5}(0) + \frac{2}{5}(1) + \frac{2}{5}(0) = 0.5 \end{aligned}$$

$$\text{Gain}(\text{income}) = 0.971 - 0.5 = 0.471$$

student	p_i	n_i	$I(p_i, n_i)$
Yes	2	0	0
No	0	3	0

$$\begin{aligned} \text{Info}_{\text{student}}(D) &= \frac{2}{5} I_{(2,0)} + \frac{3}{5} I_{(0,3)} \\ &= \frac{2}{5}(0) + \frac{3}{5}(0) = 0 \end{aligned}$$

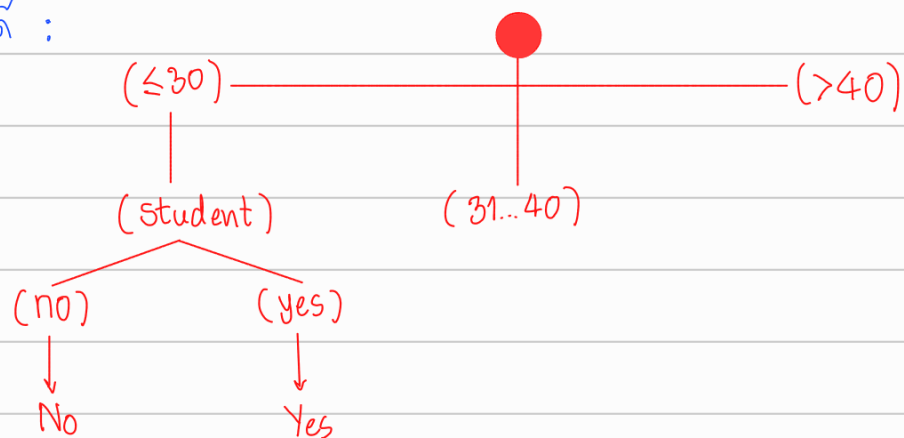
$$\text{Gain}(\text{student}) = 0.971 - 0 = 0.971 \quad \text{เลือก student}$$

credit_rating	p_i	n_i	$I(p_i, n_i)$
fair	1	2	0.918
excellent	1	1	1

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{3}{5} I_{(1,2)} + \frac{2}{5} I_{(1,1)} \\ &= \frac{3}{5}(0.918) + \frac{2}{5}(1) = 0.951 \end{aligned}$$

$$\text{Gain}(\text{credit_rating}) = 0.971 - 0.951 = 0.019$$

Decision Tree ที่ทำได้ :



>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
>40	medium	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Info}(D) = I_{(3,2)} = -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) = 0.971$$

student	p_i	n_i	$I(p_i, n_i)$
Yes	2	1	0.918
No	1	1	1

$$\begin{aligned} \text{Info}_{\text{student}}(D) &= \frac{3}{5} I_{(2,1)} + \frac{2}{5} I_{(1,1)} \\ &= \frac{3}{5} (0.918) + \frac{2}{5} (1) = 0.951 \end{aligned}$$

$$\text{Gain}(\text{student}) = 0.971 - 0.951 = 0.02$$

income	p_i	n_i	$I(p_i, n_i)$
Low	1	1	1
Medium	2	1	0.918

$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{2}{5} I_{(1,1)} + \frac{3}{5} I_{(2,1)} \\ &= \frac{2}{5} (1) + \frac{3}{5} (0.918) \end{aligned}$$

$$\text{Gain}(\text{income}) = 0.971 - 0.951 = 0.02$$

credit_rating	p_i	n_i	$I(p_i, n_i)$
fair	3	0	0
excellent	0	2	0

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{3}{5} I_{(1,2)} + \frac{2}{5} I_{(1,1)} \\ &= \frac{3}{5} (0) + \frac{2}{5} (0) = 0 \end{aligned}$$

$$\text{Gain}(\text{credit_rating}) = 0.971 - 0 = 0.971$$

เลือก Credit-rating

Decision Tree ที่เลือกไว้:

